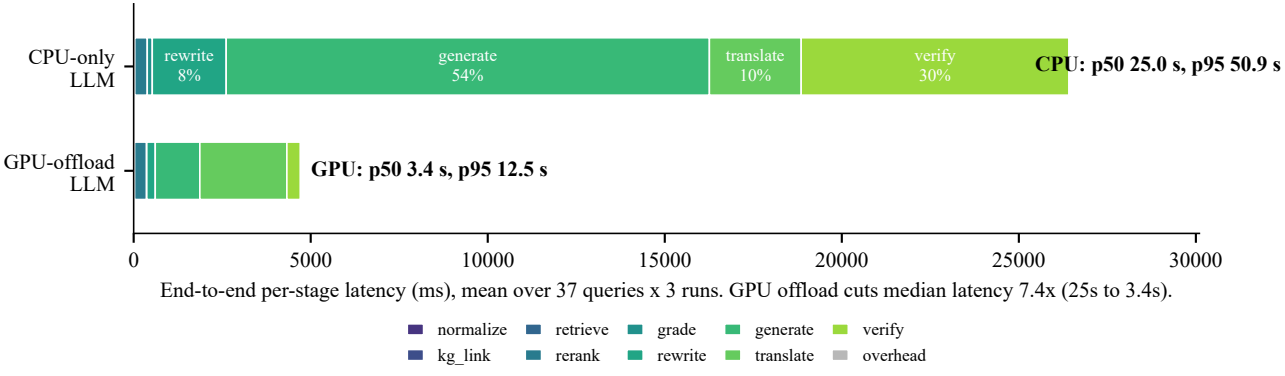
CPU-only LLM
rewrite 8%
generate 54%
translate 10%
verify 30%
CPU: p50 25.0 s, p95 50.9 s
GPU-offload LLM
GPU: p50 3.4 s, p95 12.5 s
0
5000
10000
15000
20000
25000
30000
End-to-end per-stage latency (ms), mean over 37 queries x 3 runs. GPU offload cuts median latency 7.4x (25s to 3.4s).
normalize
retrieve
grade
generate
verify
kg_link
rerank
rewrite
translate
overhead